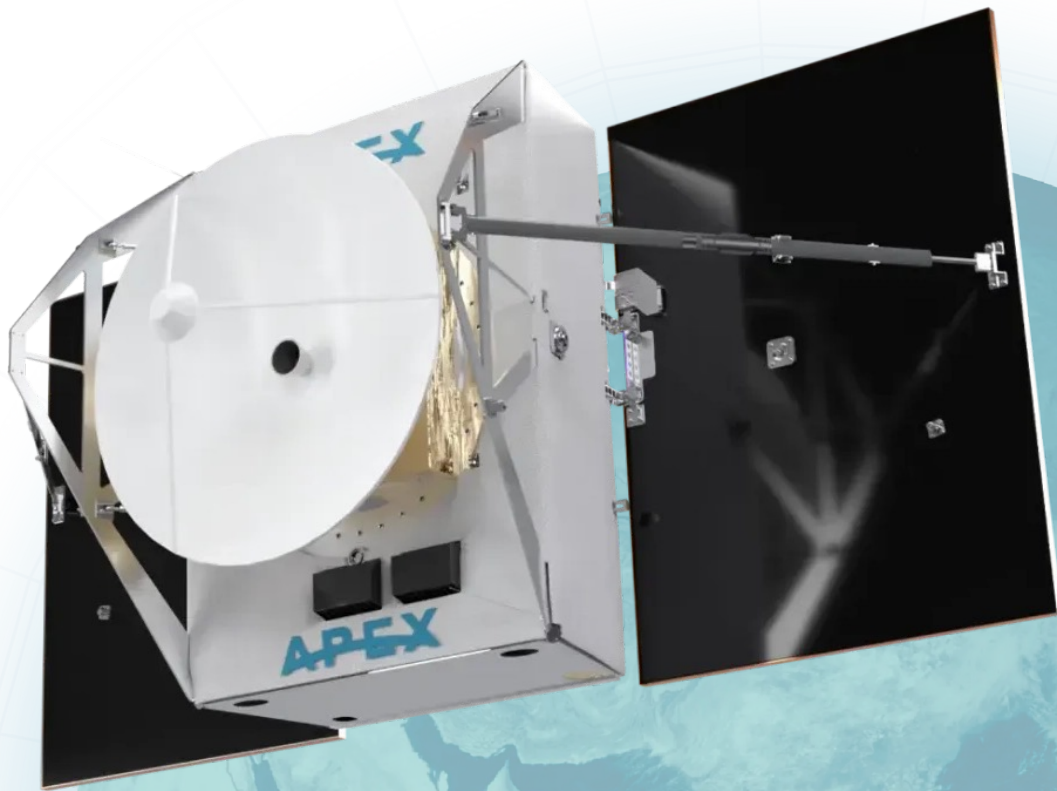


APEX

Aries User's Guide

Version 01: June 2024



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Version Control

Version Number	Date	Changes
1.0 * Beta Version	6/11/24	Initial Release

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1 Introduction

1.1 About this User's Guide

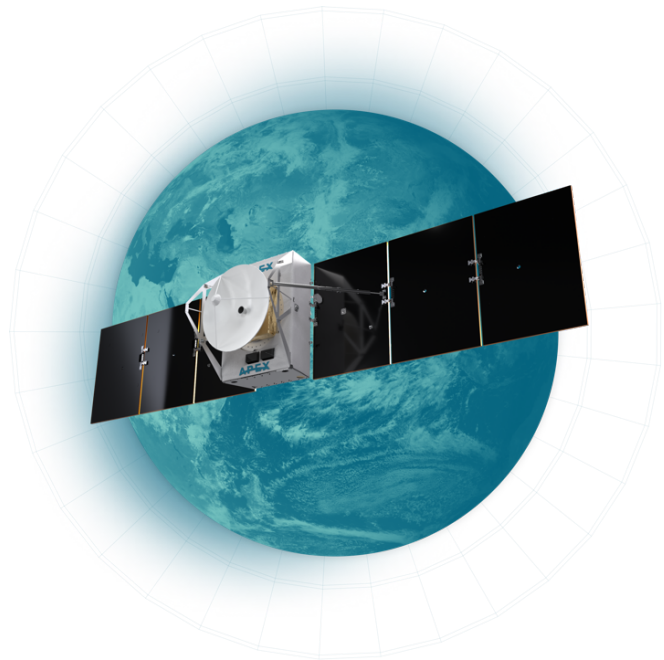
This Aries User's Guide provides an overview of Aries, one of Apex's configurable satellite bus models. It includes a summary of design, key interfaces, satellite environments, and performance specifications.

The document is provided to current and future customers as an overview with initial contracting, technical, and logistical details.

This document is intended for initial payload sizing and compatibility checks, not detailed design references. Descriptions of several Aries bus configurations are discussed, but the focus of this guide is on the Aries Base Model.

Apex reserves the right to modify, change, or update this User's Guide at any point. Detailed contractual specifications and requirements are organized with Apex on a case-by-case basis and not bound to the content included herein.

If a current or prospective customer is interested in learning more about Apex, the Aries platform, or additional services, please contact info@apexspace.com.



1.2 About Apex

Apex manufactures productized small satellite buses designed around the core principles of standardization, production at scale, and rapid delivery.

Using design-for-manufacturing approaches, Apex focuses on producing small satellite buses at scale and without mission-specific NRE. For our customers, this approach enables space missions at radically improved speed, reliability, and constellation scalability.

Apex takes a product-first approach to small satellite buses. Apex bus models can be delivered either as a base model, or with easily swappable configuration packages for different orbital regimes or performance requirements. Apex satellite bus model configurations and prices can be seen on the Apex website.¹

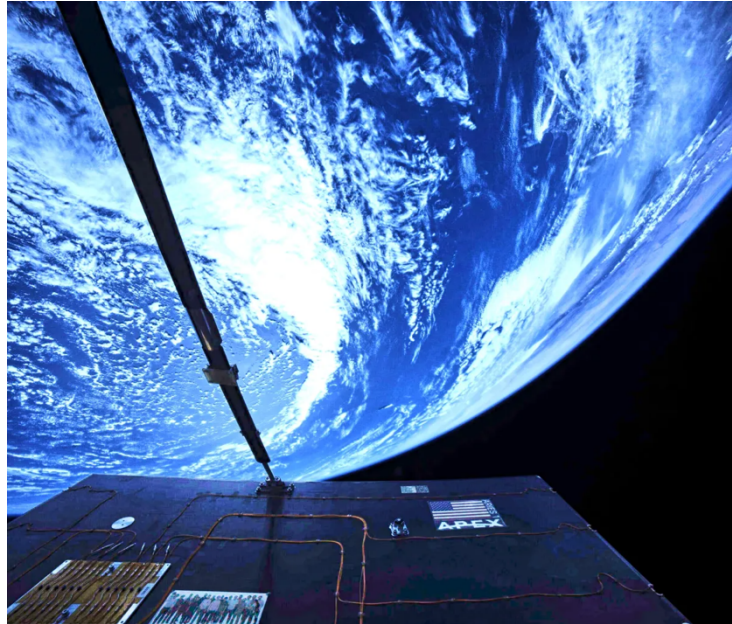


Image 1: Aries SN1 Selfie acquired March 2024

¹ <https://www.apexspace.com/order-now>

2 Aries Satellite Bus

2.1 Overview

Apex’s first bus product, Aries, is a 100 kg platform built for configurability and interoperability, providing streamlined production capabilities. Aries currently flies in Low Earth Orbit (LEO).

Aries is engineered for subsystem configurability. This means that on the production line, at the point of assembly, a common bus design can be delivered either as a Base Model, or with different configurations for power, propulsion, communication, GNC, and other factors. The Apex common bus platform can support a wide range of mission payloads without one-off, mission-specific non-recurring engineering.

Today, the Aries bus is offered in a base model or with different configuration options. Configurations enable customers to select a bus that addresses their mission requirements with no additional NRE. Outlined below are the current configuration packages for the Aries bus that are available.



Figure 1: An Aries Base Model bus shown with

	Power	SADAs	Propulsion	Encryption	Comms	GNC
Base model	Base Power Single-panel arrays; 215 W OAP*	No SADAs Fixed deployable solar arrays	No Propulsion	AES 256 Commercial encryption	Base Downlink 125 Mbps X-band	Base GNC <0.02° knowledge; <0.04° accuracy
Configuration options	Medium Power Double-panel arrays; 520 W OAP*	With SADAs Single-axis SADAs for independent sun tracking	Chemical Prop Chemical monopropellant	NSA Type 1	High-Rate Downlink 500 Mbps X-band	Precision GNC <0.0025 knowledge; <0.005° accuracy
	High Power Triple-panel arrays; 825 W OAP*		Electric Prop Xenon or Krypton HET			
			Chemical 6DOF High maneuverability			

*BOL OAP for payloads assuming a 550km circular sun-synchronous orbit with sun tracking

Figure 2: Aries configuration packages

2.2 Production

Aries buses are manufactured in serial production at Apex facilities in Los Angeles. Following automotive-type production model, Apex manufactures Aries buses to a set rate based on forecasted demand from commercial and government customers.

Depending on production slot availability, customers can receive a built and tested Aries bus with lead times far lower than current industry norms. Production volume for the Aries model is projected to reach 50+ buses per year in 2026.

Aries is compatible with most commercially available launch vehicles. Aries uses an enveloping approach to ensure environmental compliance to almost all launch vehicles.

Aries configurations are built around the standard Aries Base Model to address specific mission requirements.



Image 2: Apex's Factory One production facility in Los Angeles equipped with a ISO class 8 cleanroom

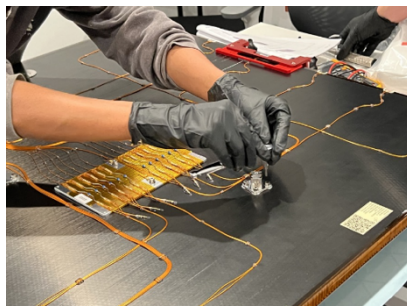
Apex currently is delivering Aries Buses off of its active production line. By Q4 2024, Apex's capacity for manufacturing is approximately 100 buses per year.



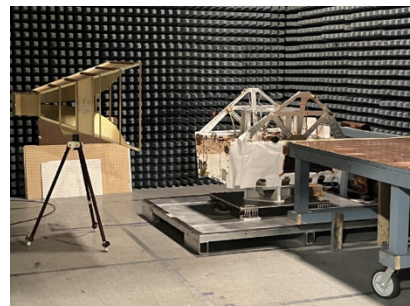
Procurement and supply chain



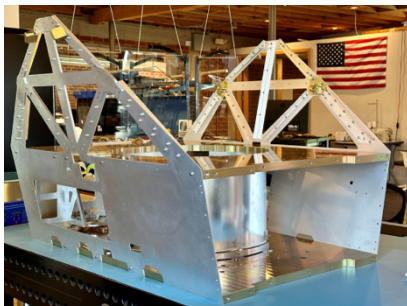
Payload integration and checkout



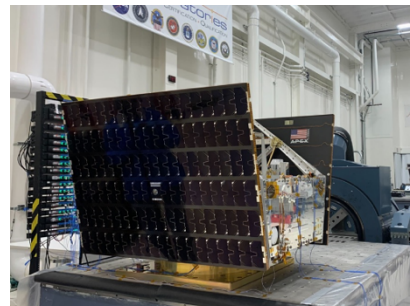
Fabrication, machining, subsystem assembly



Spacecraft environmental test



Bus structural assembly



Spacecraft environmental test



Bus integration



Inventory for future bus builds

2.3 Aries Base Model

All Aries configurations are built on the foundation of the Aries Base Model. This base configuration includes the following components outlined below in Table 1.

Component	Description
EPS System	Li-Ion battery pack and 2x deployable solar arrays, supporting 180W OAP EOL for payloads and peak power over 1kW. The Aries battery is 1680 Whr.
ADCS System	Enables 3-axis control with 4x reaction wheels, 2x star trackers, 4x sun sensors, a magnetometer, 3x torque rods, and an IMU.
RF Communications	UHF beaconing, Omnidirectional S-band TT&C, and X-band high-rate downlink.
Payload Interfaces	Flexible and configurable mechanical, thermal, electrical, and software interfaces to maximize payload interoperability.
Propulsion Systems (Optional)	Available as an add-on to the Aries Base Model. Options for monopropellant chemical or HET electrical propulsion are available.

Table 1: Aries Base Model Components

2.4 Base Model Aries Coordinate Systems

Aries uses right-handed X-Y-Z coordinate frames centered as specified below and hosts multiple coordinate systems for satellite, payload, and separation plane.

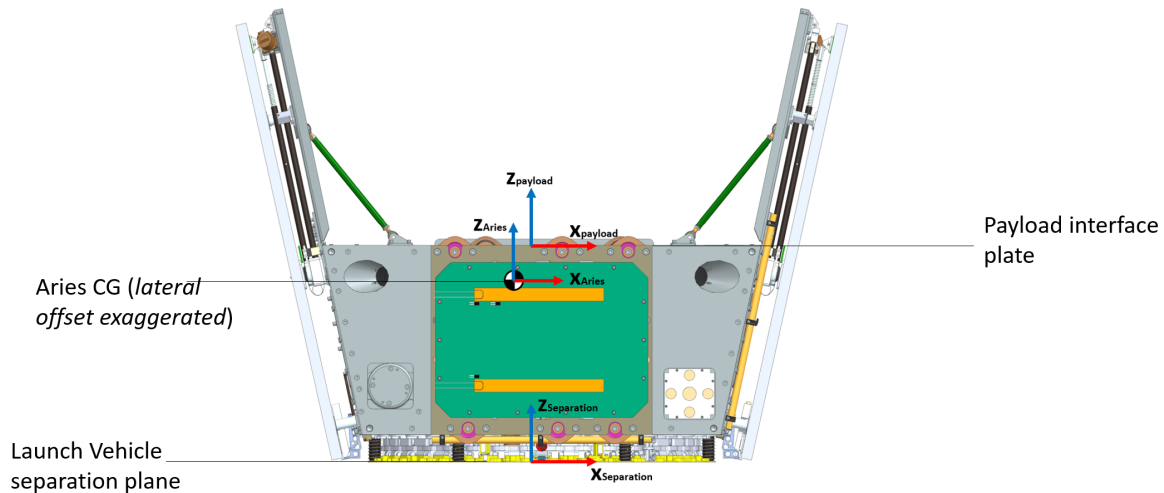


Figure 4: Aries model X-Y-Z coordinate frames.

2.4.1 Aries Satellite Coordinate Systems

The satellite coordinate system is centered on the Center of Mass of the deployed vehicle inclusive of the payload mass. This coordinate system definition is used for satellite operations and GNC control once the vehicle is on station. The +Z axis (yaw) is the payload boresight, the +X (pitch) axis is along the solar array horizontal, and the +Y axis (roll) closes the right-handed coordinate system.

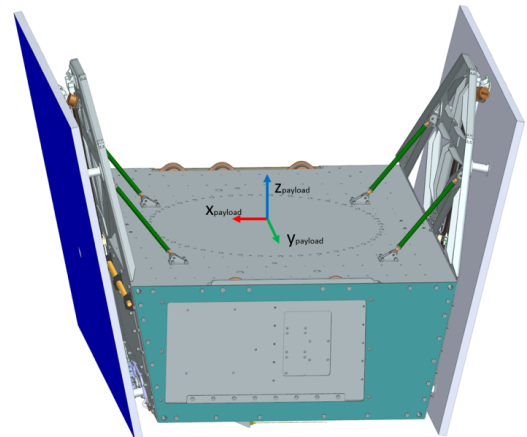


Figure 5: Aries payload interface plate

2.4.2 Payload Coordinate System

The payload coordinate system has the same axis directions as defined in the satellite coordinate system however, payloads are centered on the payload interface plate.

2.4.3 Separation Plane Coordinate System

The separation plane coordinate system has the same axis directions as defined in the satellite coordinate system.

Customers should note that the separation mechanism is centered on the platform's separation interface.

The Aries platform is compatible with numerous separation systems. Separation mechanisms should not affect the coordinate system for Aries buses.

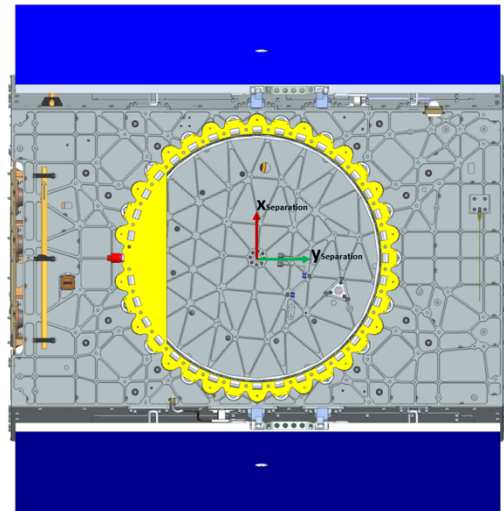


Figure 6: Separation interface on Aries

3 Base Model Aries Description

Listed below are the core subsystems and components for the Base Model Aries.

Subsystem	Key Component
Mechanical	Primary Structure
	24" Separation Mechanism
	Solar Array HDRMs (2x)
	Solar Array Hinges (2x)
Power	Solar Arrays (2x)
	Battery
	Power Control and Distribution Unit (PCDU)
Comms	X-band Transmitter
	S-band Transceiver
	UHF Transceiver
GNC	Star Trackers (2x)
	Magnetometer
	GNSS Receiver
	IMU
	Reaction Wheels (4x)
	Torque Rods (3x)
	Sun Sensors (4x)
Thermal	Heaters
	Temperature Sensors
	MLI
C&DH	Flight Computer
	Ethernet Switch

Table 2: Aries subsystem components

3.1 Aries Mechanical

The Aries Mechanical interface provides standard mounting interfaces on the +Z deck that can support payloads with masses up to 150kg. Aries is mounted to and deploys from the selected launch vehicle on a 24" separation ring.

The mechanical interface for customer payloads includes:

Interface	Description
Dia Bolt Pattern	24" diameter interface, consisting of 36x ¼-28 tapped holes
Payload Deck Grid	2" x 4" grid distributed over the deck; Aries can accommodate #8-32, #10-32 or ¼" tapped or clearance holes on this pattern

Table 3: Aries mechanical interface overview

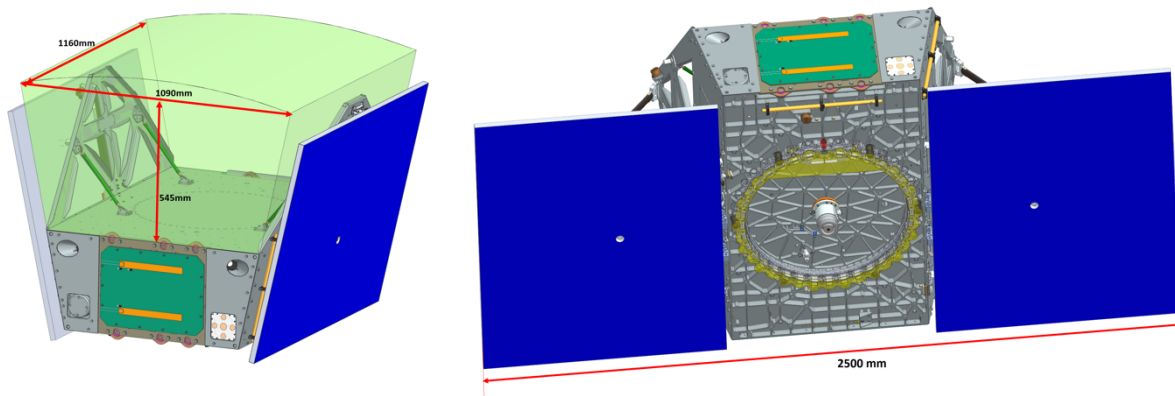


Figure 7: Stowed and deployed payload volumes on base model of Aries

3.2 Aries Thermal

The Ares-1 thermal architecture is designed to accommodate different payloads within a range of possible environments. Rather than approach each payload accommodation as a point design, the Payload user is presented with an acceptable thermal envelope based solely on orbital altitude and maximum expected orbital Beta Angle (angle between orbital plane and solar vector).

If payload parameters fall outside of this range, further analysis may be required to show compatibility. Apex recognizes that the most efficient designs with respect to mass and complexity thermally couple the Payload and Bus. The intent of this table is to show that thermal isolation between the payload and bus is not required, and that Apex will work with each customer as necessary to show compatibility.

Payload Thermal Accomodation Capability							
Altitude [km]	Payload Dissipation on Top Deck [W]						
1000	100	100	100	80	60	60	60
900	100	100	100	80	60	60	60
800	100	100	100	80	60	40	40
700	100	100	100	80	60	40	40
600	100	100	100	80	80	40	40
500	100	100	100	80	80	40	40
400	100	80	80	80	80	40	40
	0	15	30	45	60	75	90
Maximum Beta Angle [deg]							

Table 4: Aries Thermal Accommodation

The Aries platform provides 18 temperature sensor channels along with 18W, 2A heater circuits. Additional temperature sensors and heaters can be made available to customers upon request.

3.3 Aries Power

Aries's Power Control and Distribution Unit (PCDU) regulates power distribution to all components on the spacecraft including the payload.

Nominally, a payload can pull 2 amps per channel. During an extended transient or short circuit event, the load switch has internal protection to turn off a channel.

Parameter	Value	Unit
Base Aries Solar Array Power Generation (EOL)	450	W
Energy Storage	1680	Whr

Table 5: Aries power interface parameters

3.4 Aries Communications

The Aries bus flies with a redundant comms system in UHF, S-band, and X-band frequencies. The UHF and S-band radios are omnidirectional with a directional X-band communication system for high-data-rate applications.

Standard Aries models are encrypted with AES-256. Aries is compatible with a limited set of NSA Type-1 encryption units to support customer data security needs.

Frequency	Peak Data Rate
UHF	Bidirectional: 19.2 kbps
S-band	Uplink: 750 kbps Downlink: 3.5 Mbps
X-band	Downlink: 125 Mbps

Table 6: Radios hosted on the Aries satellite bus with respective peak data rates

3.5 Aries Command and Data Handling

The Aries spacecraft provides multiple communications and data storage capabilities between the user's payload and bus. The interfaces are stated in the table below. The interfaces are available during integration and test phases as well as mission operations.

Interface	Qty
Payload High Speed Data Interface - LVDS	1
Payload High Speed Data Interface - Ethernet	4
Payload Command and Telemetry Interface	3
Payload Discrete	5
Payload PPS	1

Table 7: Bus-payload data interfaces

3.6 Aries Software

Aries's payload software interface allows payloads to receive bus telemetry and send bus commands. Apex provides a sequencing language which can be used to automate routine tasks. Customers interested in operating an Aries platform also have access to the Aries command library or the Apex Virtual Flatsat.

3.7 Aries Guidance, Navigation, and Control

3.7.1 GNC Performance

Capability	Specification
Pointing Knowledge	0.06 deg 3-sigma
Pointing Control	0.12 deg 3-sigma
Slew Agility	< 2 deg/sec roll/pitch/yaw
Position/Velocity/Time Accuracy	10 m / 0.03 m/s / 40 ns rms

Table 8: Aries Base GNC configuration capabilities

4 Working with Apex

4.1 Standard Deliverables

The following table summarizes the deliverable items and services to be provided by Apex. Please note: These deliverables represent Apex's standard deliverables. As required, this list may be modified for customer and/or mission-unique needs.

Deliverable	Deliverable Type	Description
Satellite Bus	Hardware	Satellite bus as described in this document
Virtual Flatsat	Software	Bus-to-Payload interface for payload development
Bus Emulator	Hardware	Desktop unit emulating bus software payload interfaces
Bus Shipping Container	Hardware	Shipping container for bus transportation
Bus Design Description	Document	Design description of the bus containing design and performance content, including payload interfaces. Provided in lieu of traditional bus design reviews
Bus Comprehensive Functional Test (CFT) Report	Document	Report certifying that bus AIT is complete and successful, and that the bus is ready for delivery and payload integration
Bus Functional Test Procedure	Document	Procedures for satellite bus functional and software checkout
Bus Launch Integration Procedure	Document	Procedures for bus loading onto and integration with the launch vehicle
Bus On-Orbit Handbook	Document	Bus operational handbook, TM/TC dictionary, and SGICD
Project Status Updates (Monthly)	Document / Meeting	Project schedule and technical progress updates

Table 9: Apex's standard deliverable table

5 Appendix

5.1 Acronyms

Term	Definition
NRE	Non-Recurring Engineering
GNC	Guidance, Navigation, and Control
SADA	Solar Array Drive Assembly
OAP	Orbit Average Power
EPS	Electrical Power System
ADCS	Attitude Determination and Control System
RF	Radio Frequency
HET	Hall Effect Thruster
HDRM	Attitude Determination and Control System
GNSS	Global Navigation Satellite System
IMU	Inertial Measurement Unit
MLI	Multi-Layer Insulation

Table 4: List of acronyms